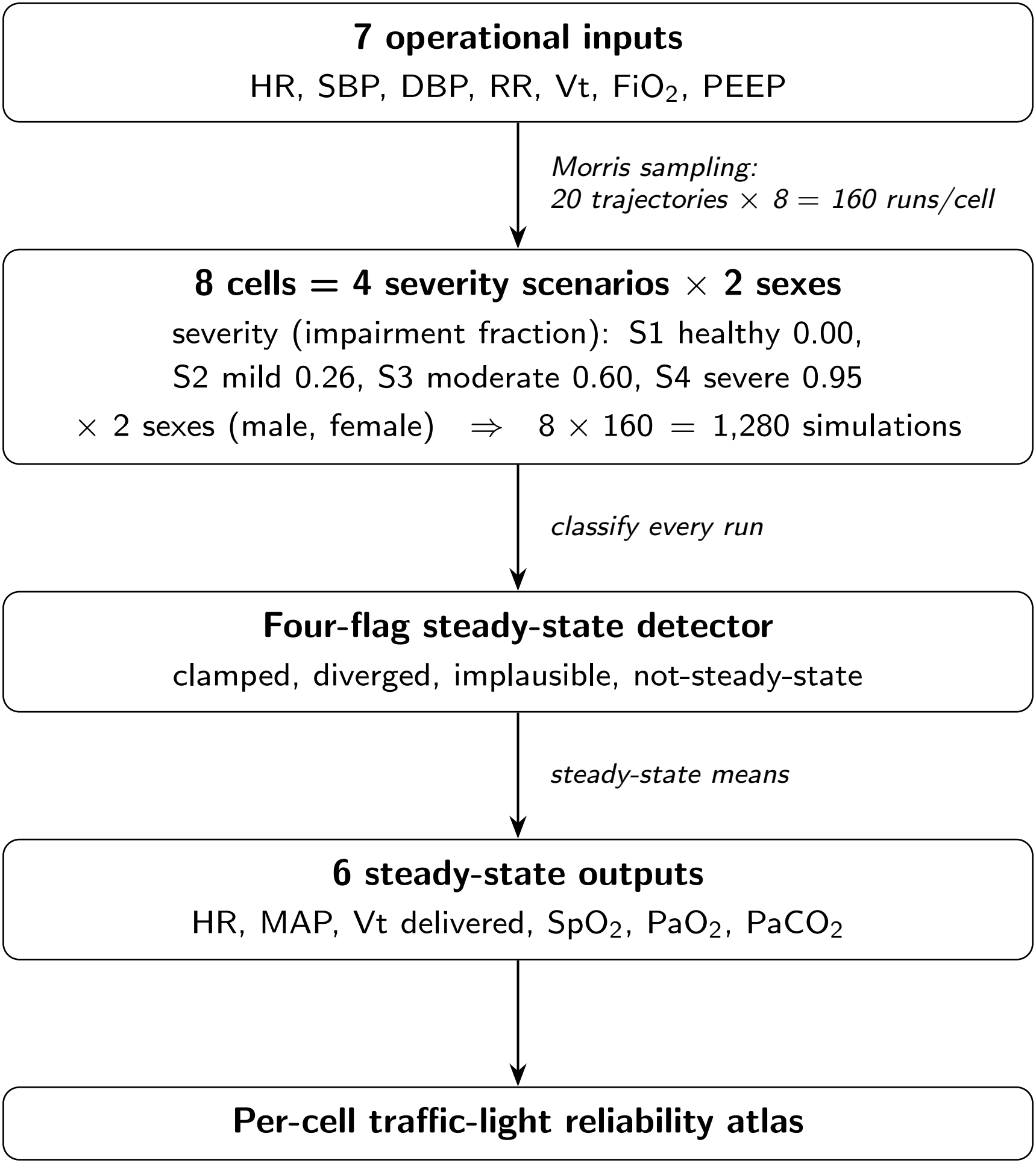

7 operational inputs
HR, SBP, DBP, RR, Vt, FiO2, PEEP
Morris sampling:
20 trajectories × 8 = 160 runs/cell
8 cells = 4 severity scenarios × 2 sexes
severity (impairment fraction): S1 healthy 0.00,
S2 mild 0.26, S3 moderate 0.60, S4 severe 0.95
× 2 sexes (male, female) ⇒ 8 × 160 = 1,280 simulations
classify every run
Four-flag steady-state detector
clamped, diverged, implausible, not-steady-state
steady-state means
6 steady-state outputs
HR, MAP, Vt delivered, SpO2, PaO2, PaCO2
Per-cell traffic-light reliability atlas